

# Katravath Srinu

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## Education

**Indian Institute of Technology (BHU), Varanasi**

*Expected 2027*

*B.Tech in Computer Science and Engineering*

- *Relevant Coursework:* Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Generative AI Fundamentals, Probability & Statistics, DBMS, Operating Systems, Computer Networks

## Technical Skills

**Programming Languages:** Python (Primary), C++, JavaScript, SQL

**AI / ML:** LLMs, LangChain, RAG, NLP, Scikit-learn, Prompt Engineering, Generative AI

**Large-Scale Data Analytics:** Pinecone (Vector DB), MongoDB, MySQL, Data Pipeline Design

**Software Development:** REST APIs, WebSockets, Node.js, Express.js, System Design, CI/CD

**Cloud / Infrastructure:** AWS (EC2, Lambda, API Gateway, S3), Docker

**Tools:** Git, Linux, Postman

**Soft Skills:** Problem Solving, Cross-functional Collaboration, Analytical Thinking, Client Communication

## Experience

**Backend Open Source Contributor**

*2025 – 2026*

*Independent / Remote*

- Contributed **150+ pull requests** across the full software development life cycle — from algorithm design to implementation and review — applying industry-standard engineering practices.
- Optimized computational efficiency of API response handling logic across multiple endpoints, reducing latency and improving performance through rigorous algorithmic analysis.

## Projects

**Financial Document QA System (RAG Pipeline) | *Python, AWS, LangChain, GPT, Pinecone*** *2025*

- Built an end-to-end generative AI pipeline to extract and summarize insights from financial reports using RAG with Pinecone vector database, enabling accurate large-scale document querying.
- Deployed serverless inference pipeline on **AWS Lambda + API Gateway** for scalable, production-ready AI querying — demonstrating full analytics value-chain ownership from design to deployment.

**VoIP Attack Detection System | *Python, SIP, PCAP, Wireshark***

*2025*

- Developed an intrusion detection pipeline achieving **94% attack classification accuracy** by analyzing 10+ GB of PCAP traffic traces across multiple attack scenarios.
- Engineered packet-level feature extraction reducing false positives by **40%** — demonstrating large-scale data analytics and algorithm optimization skills core to BCG X.

**Real-Time Chat Application | *Node.js, Express.js, WebSockets, MongoDB***

*2025*

- Built a scalable multi-user messaging platform with WebSocket-based low-latency architecture supporting concurrent client communication.
- Implemented authentication workflow and persistent message storage backend, applying robust full-cycle software development practices.

## Certifications & Achievements

**HackerRank Software Engineer Intern Role Certification**

*Jan 2026*

- Certified proficiency in Data Structures, Algorithms, SQL, and Python — BCG X's primary required programming language.
- Solved **250+ DSA problems** on LeetCode and Codeforces (Max Rating: 1200) — strong algorithmic and computational problem-solving foundation.
- Secured **AIR 11227** in JEE Advanced 2023 among 150,000+ candidates; selected for B.Tech CSE at IIT (BHU).

## Research & Thought Leadership Interests

- **Generative AI & LLMs:** Actively exploring RAG, prompt engineering, and LLM fine-tuning for real-world enterprise applications.
- **AI for Social Impact:** Keen interest in applying AI to climate change, financial inclusion, and large-scale societal challenges — aligned with BCG X's mission areas.